

Introduction to Healthcare and Health Insurance

Introduction

Healthcare is the system of doctors, hospitals, clinics, medicines and treatments that help people stay healthy or recover from illness. Health insurance is a financial protection plan that helps pay medical expenses, so patients do not have to bear the full cost.

1. Basic Definitions

Healthcare: Organized medical services including doctors, hospitals, tests, surgeries, and preventive care.

Health Insurance: A policy that shares medical costs between the patient and the insurance company.

Access to Healthcare: People should be able to receive affordable and timely treatment whenever they need it.

2. What Health Insurance Covers

- Hospital stays and surgeries
- Primary care visits and health check-ups
- Prescription medicines
- Emergency room visits and ambulance services

3. Types of Health Insurance in the USA

Employer-Sponsored Insurance: Given through employers and covers nearly half (49%) of Americans.

Individual/Family Plans: Purchased directly through the ACA Marketplace or private insurers. It covers around 6% of the population.

Government Insurance: Medicare, Medicaid and CHIP for eligible people.

Uninsured: Around 8–9% of people have no health insurance.

4. Government Programs

Medicaid: For low-income families, pregnant women, children, disabled people and some seniors.

Medicare: Federal insurance mainly for people aged 65+, certain disabled people and ESRD/ALS patients. Parts A, B, C, and D cover hospital care, medical services and medicines.

CHIP: Covers children whose families cannot afford private insurance but do not qualify for Medicaid.

5. Affordable Care Act (ACA)

Passed in 2010 to improve access to health insurance. It created the Health Insurance Marketplace, provides subsidies, protects people with pre-existing conditions, and allows states to expand Medicaid.

6. Stakeholders

Patients: Receive healthcare services.

Providers : Doctors, hospitals, and clinics that provide treatment.

Payers : Insurance companies and government programs that process and pay eligible healthcare claims.

Policymakers : Create healthcare laws, regulations, and programs.

Industry suppliers: Provide medicines, medical devices, and healthcare equipment.

7. Healthcare Flow

Patient → Provider → Insurance/Payer → Claim Processing → Approval/Denial → Payment

- A patient receives treatment from a healthcare provider. The provider submits a claim to the insurance company.
- The insurer reviews the claim based on the patient's coverage and then approves or denies it. If approved, payment is processed according to the insurance plan.

End To End Payers Workflow

1. Member Enrollment (834)

- Member details such as demographics, plan information, and eligibility are received.
- Enrollment data is loaded into the payer system through 834 transactions.
- The system uses this information to verify member coverage during healthcare services and claims processing.

2. Provider Network Development

- Healthcare providers such as doctors, hospitals, and clinics are contracted with the payer.
- Provider details, contracts, and reimbursement rates are maintained in the system.
- The network information is used during claims processing to determine provider eligibility and payment.

3. Member Services

- Members contact the service team for benefit, eligibility, and coverage-related questions.
- Support is provided for ID cards, claims status, and plan information.
- Member issues and requests are tracked and resolved.

4. Prior Authorization

- Providers submit requests for services that require approval before treatment.
- The payer reviews the request based on benefits and healthcare guidelines.
- The service is approved or denied based on the review outcome.

5. Healthcare Service Delivery

- The member receives treatment from a healthcare provider.
- The provider verifies member eligibility and coverage details.
- Service details such as diagnosis and procedures are recorded.

6. Claims Submission (837)

- The provider creates and submits an electronic claim after providing services.
- The claim is sent through a clearinghouse for validation.
- The validated claim is routed to the payer system for processing.

7. Claims Adjudication

- The payer system reviews the claim details.
- Eligibility, benefits, provider contracts, coding, and pricing are verified.
- The claim is approved, partially paid, or denied based on rules.

8. Payment & Remittance (835)

- Payment is generated for approved claims.
- An electronic remittance file is sent to the provider.

- The remittance explains payment details, adjustments, and denial reasons.

9. Denials & Appeals

- Denied claims are reviewed to identify the reason for rejection.
- Providers can correct errors and resubmit claims.
- Appeals are submitted with supporting documents when required.

10. Compliance & Reporting

- Operational and regulatory reports are generated.
- Data is monitored to ensure healthcare rules and requirements are followed.
- Reports support audits and business decisions.

11. Grievances & Appeals

- Member complaints related to services or coverage are received.
- Issues are reviewed and investigated.
- A resolution is provided based on policies and guidelines.

QNXT Architecture

QNXT Architecture is the overall structure of the QNXT healthcare system. It shows how different components work together to manage health insurance operations such as member enrollment, provider management, claims processing, benefits, billing, and payments.

- It receives member details (enrollment and eligibility information) and stores coverage details.
- It maintains provider information like contracts, networks, and payment rules.
- It receives claims from providers after healthcare services are provided.
- It checks claims by verifying:
 - Is the member eligible?
 - Is the service covered?
- Based on these checks, QNXT approves, denies, or adjusts claims.
- For approved claims, it calculates payment amounts and sends payment details.
- It also supports reporting, compliance, and integration with other healthcare systems.

Key EDI Transactions

834 - Member Enrollment

837 - Claim Submission

835 - Payment/Remittance

997 - Functional Acknowledgement

Electronic Data Interchange

HIPAA (Health Insurance Portability and Accountability Act) is a U.S. federal law enacted in **1996** to protect the privacy and security of patients' health information.

The main purpose of HIPAA is to ensure that healthcare organizations and their business partners handle patient information securely and confidentially.

Simple Definition

HIPAA is a law that tells healthcare organizations:

- Keep patient information private.
- Share health information only with authorized people and protect patient data from theft, misuse, or unauthorized access.

Protected Health Information (PHI).

It refers to any information that can identify a patient and is related to their healthcare.

Examples of PHI: Patient Name, Address, Phone Number, Email Address etc...

Example:

Patient Name: John Smith

Diagnosis: Diabetes

Insurance ID: ABC12345

This is PHI because it identifies a patient and contains medical information.

What is EDI?

EDI stands for **Electronic Data Interchange**.

It is the electronic exchange of business documents between two computer systems in a standardized format without manual intervention.

Instead of printing forms, scanning them, or emailing PDFs, computers communicate directly using EDI.

Simple Example

Without EDI:

Hospital → prints claim → emails insurer → insurer manually enters data

With EDI:

Hospital System → sends electronic claim → Insurance System receives automatically

No human typing is required.

- Process claims faster
- Improve communication

EDI Standards

EDI standards define how electronic documents should be structured. They ensure every organization sends and receives information in the same format.

Common EDI Standards

1. ANSI ASC X12: Most commonly used in the United States.

Used for:

- Healthcare
- Insurance
- Finance & Government

2. EDIFACT: Developed by the United Nations. Mostly used outside North America.

3. NCPDP: National Council for Prescription Drug Programs.

Used specifically for:

- Pharmacies & Drug claims
- Prescription processing

4. HL7: Health Level Seven: It is Used to exchange clinical information such as:

- Lab reports
- Medical records
- Patient admissions

Unlike X12, which focuses mainly on insurance transactions, HL7 is used for clinical data exchange.

EDI File Structure – How It Works

1. Interchange Control Header (ISA)

- This is the beginning of an EDI file.
- It identifies the sender and receiver information.
- It contains control numbers and communication details.

2. Functional Group Header (GS)

- Groups similar transactions together within an interchange.
- Contains information about the type of transaction and sender/receiver details.
- Helps systems identify and process related transactions.

3. Transaction Set Header (ST)

- Marks the beginning of an individual transaction.
- Identifies the specific transaction type being sent.

4. EDI Payload Segments

- Contains the actual business data being exchanged.
- Includes details such as Member information, Provider information, Claim details.
- The data is arranged in segments based on HIPAA standards.

5. Transaction Set Trailer (SE)

- Marks the end of a transaction.
- Contains transaction control numbers.
- Ensures the transaction was received completely and accurately.

6. Functional Group Trailer (GE)

- Marks the end of a functional group.
- Provides the count of transactions included in the group.
- Helps validate the completeness of the group.

7. Interchange Control Trailer (IEA)

- Marks the end of the complete EDI interchange.
- Confirms the number of functional groups and control numbers.
- Ensures the entire file structure is complete.

Types of Healthcare EDI Transactions

1. 834 – Enrolments Transaction
2. 837 – Claim Transaction
3. 835 – Payment/Remittance Transaction
4. 270/271 – Eligibility Transaction
5. 276/277 – Claim Status Transaction
6. 278 – Prior Authorization Transaction
7. 820 – Premium Payment Transaction

SNIP Level Edits

SNIP stands for **Strategic National Implementation Process**.

It is a framework developed to validate HIPAA EDI transactions and categorize errors.

Its purpose is to ensure transactions comply with HIPAA standards before they are processed.

Why SNIP Exists??

Healthcare EDI files contain hundreds of data elements. If any required field is missing or incorrectly formatted, the transaction may fail. SNIP helps detect these issues early.

Seven SNIP Levels

Type 1 – Syntax Integrity

Checks whether the EDI file follows the correct format.

Example:

Missing segment terminator (~).

Type 2 – HIPAA Syntax Validation

Checks compliance with HIPAA implementation guides.

Example: Required fields must be present.

Type 3 – Balancing

Verifies totals and counts.

Example: The number of claims matches the number specified in the trailer.

Type 4 – Situation Testing

Checks conditional business rules.

Example:

If a patient is female, pregnancy information may be required.

Type 5 – External Code Set Validation: Validates standard medical codes.

Examples:

- ICD Diagnosis Codes
- CPT Procedure Codes
- HCPCS Codes

Type 6 – Product Type or Line of Service Validation

Ensures services match the member's insurance plan.

Example: Dental claims should not be billed under a medical-only plan.

Type 7 – Trading Partner Validation

Checks payer-specific rules.

Example: Molina Healthcare may require additional fields beyond the standard HIPAA requirements.

Medicaid Enrollment Process Flow

State Agency Process (Zone 1):

The State Medicaid Agency is responsible for enrolling eligible members into a Medicaid health plan. Here is the process flow:

Step 1: Award Contract to Payers

The State Medicaid Agency signs contracts with managed care organizations (MCOs) such as Molina Healthcare.

Step 2: Review Application (APP)

The applicant submits a Medicaid application with personal and financial information.

The state reviews Name, Date of Birth, Address, Income & verify whether the application is complete and ready for eligibility review.

Step 3: Determine Eligibility (ELIG): The State Agency checks whether the applicant qualifies for Medicaid. Eligible Continue to plan selection else denied.

Step 4: Plan Selection (PLAN)

Once approved, the member chooses a health plan.

Step 5: Auto Assignment (AUTO)

If the member does not choose a plan within the allowed time, the State Agency automatically assigns one. The assignment is based on Geographic location and Provider availability.

Step 6: Send EDI 834 Enrollment File

After planning selection or auto-assignment, the State Agency creates an EDI 834 Enrollment file and sends it to the selected payer (such as Molina).

The 834 file contains: Member ID, Member Name, Date of Birth, Gender, Health Plan Information, Enrollment Status, etc.

ZONE-2: Payer Enrollment Processing

- After the State Agency sends the EDI 834 Enrollment file, Molina receives and processes it.
- The system validates the enrollment data, creates a member record, assigns a (PCP), welcomes the member, and shares information with healthcare vendors.

Step 1: Process 834 File (Intake & Validate)

The first step is to receive the EDI 834 Enrollment file from the State Agency.

The system validates the file by checking if the File format is correct.

* Required member information is present.

* No duplicate enrollment exists.

Step 2: Create Enrollment (MBR ID)

After validation, Molina creates a Member Record in the QNXT system. A unique Member ID (MBR ID) is generated for each member. To create the member's profile in the payer system.

Step 3: Assign PCP (Primary Care Provider)

Once the member record is created, Molina assigns a Primary Care Provider (PCP). A PCP may be Family Doctor, General Physician, Internal Medicine Doctor, Pediatrician. If the member selected a PCP during enrollment, that provider is assigned. Otherwise, Molina assigns one based on: Member's location, Provider availability, Network participation.

Step 4: Member Outreach (Welcome & Assignment)

After the PCP assignment, Molina contacts the new member. To inform the members about their enrollment and available healthcare services.

Step 5: Send Enrollment Data to Downstream Systems

Finally, Molina sends enrollment information to other connected healthcare systems and vendors.

Reconciliation Process: It is having follow steps:

1. 834 Updates (Daily Enrollment Updates): Member enrollment changes are received daily through 834 files. These updates include new enrollments, demographic changes, and corrections, which are processed and updated in the system.

2. Monthly Reconciliation

Enrollment data is compared between payer and external systems to identify any differences. Any mismatches are reviewed and resolved to maintain accurate member records.

3. Annual Redetermination

Member eligibility is reviewed annually to confirm whether members still qualify for coverage. Based on the review, members are renewed, modified, or terminated.

4. Plan Changes

Members can change their health plans during the enrollment period. The selected plan changes are processed and updated in the system.

5. Disenrollment

Members are removed from the health plan when they lose eligibility, request termination, or move out of the coverage area. The termination details are updated accordingly.

6. Capitation (PMPM Payment)

Capitation payments are calculated based on the number of enrolled members. The payer receives a fixed Per Member Per Month (PMPM) payment according to enrollment data.

Claims and Encounters

A claim is a request submitted by a healthcare provider to the payer for payment after providing medical services to a member.

It contains details about the patient, provider, services provided, diagnosis, procedures, and charges.

Inbound Claim Processing

Step 1: Inbound Claim Processing

- Providers submit claims either as paper claims or electronic (EDI 837) claims.
- Paper claims are scanned using OCR and converted into electronic format.
- The File Transfer System (FTS) validates the files, checks for duplicate claims, and verifies the sender.
- A 997 Acknowledgement is sent to confirm that the claim file has been received successfully.

Step 2: Member & Provider Match

- The system matches the claim with the correct member and provider using details such as Member ID, Name, and Date of Birth.
- If a valid match is found, the claim continues to be processed.
- If no match is found, the claim is sent to the **Claims Editor**.

Step 3: Claims Routing

- After matching, the claim is routed to the appropriate environment:
 - **Medicare** – Active Medicare member.
 - **Medicaid** – Active Medicaid member.
 - **Claims Editor** – No member/provider match or inactive enrollment.

Step 4: Claims Editor

- Claims that fail automatic validation are manually reviewed.
- The examiner corrects missing or incorrect information.
- Corrected claims are loaded into QNXT for further processing.
- If the claim cannot be corrected, a denial notice is sent to the provider.

Step 5: Claims Processing

- The claim undergoes initial validation to check eligibility, required fields, and coding.
- It then enters automated adjudication, where business rules determine whether it should be approved, denied, or sent for manual review.
- Simple claims are automatically approved (Autopay).
- Complex claims are pending for manual review by a claim examiner.

Step 6: Final Decision

- **Approved Claim:** Payment is calculated and sent to the provider along with an 835 Electronic Remittance Advice (ERA).
- **Denied Claim:** A denial notice explaining the reason is sent to the provider.
- After payment or denial, the claim processing is completed.

Inbound Encounter Processing: (provider to payer)

Inbound Encounter Processing receives encounter files from providers and validates them before loading them into the system. Ensure that only valid encounter records are stored and processed.

process Flow

1. **Receive Encounter File (837 I/P/D)** from providers/vendors.
1. **HIPAA Validation** – Check HIPAA 5010 compliance.
 - a. **Fail** → Sent to Non-HIPAA process.
 - b. **Pass** → Loaded into the staging database.
1. **Member Match** – Verify the member that exists.
2. **Provider Match** – Verify the provider exists.
3. **Validation Rule Engine** – Applies business rules, duplicate check, and void/replace validation.
4. **Editor Data Fix** – Claims with errors are corrected manually.
5. **ODS (Operational Data Store)** – Valid encounters are stored.
6. **277CA Notification** – Providers receive acceptance or rejection status.

Outbound Encounter Processing:

Outbound Encounter Processing prepares validated encounter records and submits them to the state or federal agency.

process Flow:

- Extract encounter data from ODS (Operational Data Store) and Staging tables.
- Check the encounter status to identify whether it is:

Original/New encounter, Replace, Cancelled/Void encounter

- Perform additional validation to verify that the encounter data is complete, accurate, and follows the required business rules.
- If validation fails, send the encounter to CEMS/Encounter Management for error review and correction.
- If validation passes, mark the encounter as "Submitted."
- Generate the 837-encounter file based on the type of service:
 - 837P – Professional service
 - 837I – Institutional services
- Submit the 837 files to the State Medicaid Agency/CMS or concerned agency.
- The Agency validates the submitted file and sends a response.
- The encounter is marked as Accepted or Rejected based on the agency's response.
- Store the submission and response details in the database/data lake.
- Generate reports and monitor the encounter status, including accepted, rejected and error encounters.

Member Portal

Member Portal – Introduction:

A Member Portal is a secure online platform that allows health plan members to access healthcare information and manage their services digitally.

Purpose of Member Portal

- Provide members with easy access to their health plan information.
- Enable self-service options and reduce dependency on customer support.
- Improve member engagement and experience.

Key Features

Member Profile Management – View and update personal information.

Eligibility & Benefits – Check coverage details, plan benefits, and eligibility status.

Claims Information – View claim status and claim history.

Provider Search – Find network providers and healthcare facilities.

ID Card Access – View and download digital member ID cards.

Communication – Receive important health plan notifications and messages.

Documents & Forms – Access to plan documents, statements, and forms.

Benefits

- 24/7 access to health plan services.
- Faster and easier information access.
- Improved communication between members and health plans.
- Supports better healthcare decision-making.

Provider Portal

Provider Portal

- The Provider Portal is an online platform used by healthcare providers to interact with the health insurance payer.
- Providers can use the portal to check member eligibility, benefits, prior authorizations, claims, remittances, and overpayments.

Eligibility and Benefits

- Used by providers to check whether a patient/member is eligible for healthcare coverage.
- Providers can view the members: Member ID, Date of Birth, Payer information, Plan and coverage dates, Health benefit plan.
- **Purpose:** To confirm the patient's insurance coverage and understand what services are covered.

Availity Provider Portal

- Availity is a multi-payer provider portal that allows healthcare providers to interact with multiple insurance payers through a single platform.
- Availity mainly provides **two types** of functionalities:
 - Availity Native Functionalities
 - Availity Payer Spaces
- 1. Availity Native Functionalities**
- Availity provides common or "Native" functionalities that are available to providers across participating payers.
- The major Native functionalities include:

Eligibility and Benefits

Prior Authorizations

Claims and Remittance Management

2. Availity Payer Spaces

- Payer Spaces are payer-specific areas available within the Availity Portal.
- They provide applications and functionalities specific to a particular payer.
- They can provide Single Sign-On (SSO) with legacy provider portals, allowing providers to access certain legacy applications without having to log in separately.

Prior Authorization

Prior Authorization means getting approval from the health insurance payer before providing certain medical services.

- Providers can submit an authorization request through the portal.
- The provider enters patient and service-related information.
- The request receives a reference/certification number.
- The authorization can have different statuses, such as pending, confirmed other status depends on review process.

Purpose: To verify whether a particular medical service requires authorization and track its approval status.

Claims and Remittances

- This section helps providers manage and track claims and payments.
- Main features include:
 - Claim Status Inquiry – Check the current status of a submitted claim.
 - Remittance Viewer – View payment/remittance information related to claims.
 - Appeals – Submit or manage appeals for claim decisions.

Claim Status Inquiry: Used by providers to search and check the status of a claim.

Purpose: To find out whether a claim is submitted, processed, paid, denied, or in another status.